

EXPERIMENT 37

Authentication Password-Failure Analysis

AIM

To identify repeated password failures against one account and failures distributed across multiple accounts from the same source.

ALGORITHM

1. Read authentication records.
2. Count failures by username and source IP.
3. Detect repeated failures against one account.
4. Detect one source attacking multiple accounts.
5. Display separate findings.

CODE

```
from collections import Counter, defaultdict

logs = [
    ("10:00", "alice", "10.0.0.5", "FAIL"),
    ("10:01", "alice", "10.0.0.5", "FAIL"),
    ("10:02", "alice", "10.0.0.5", "FAIL"),
    ("10:03", "bob", "10.0.0.5", "FAIL"),
    ("10:04", "carol", "10.0.0.5", "FAIL")
]

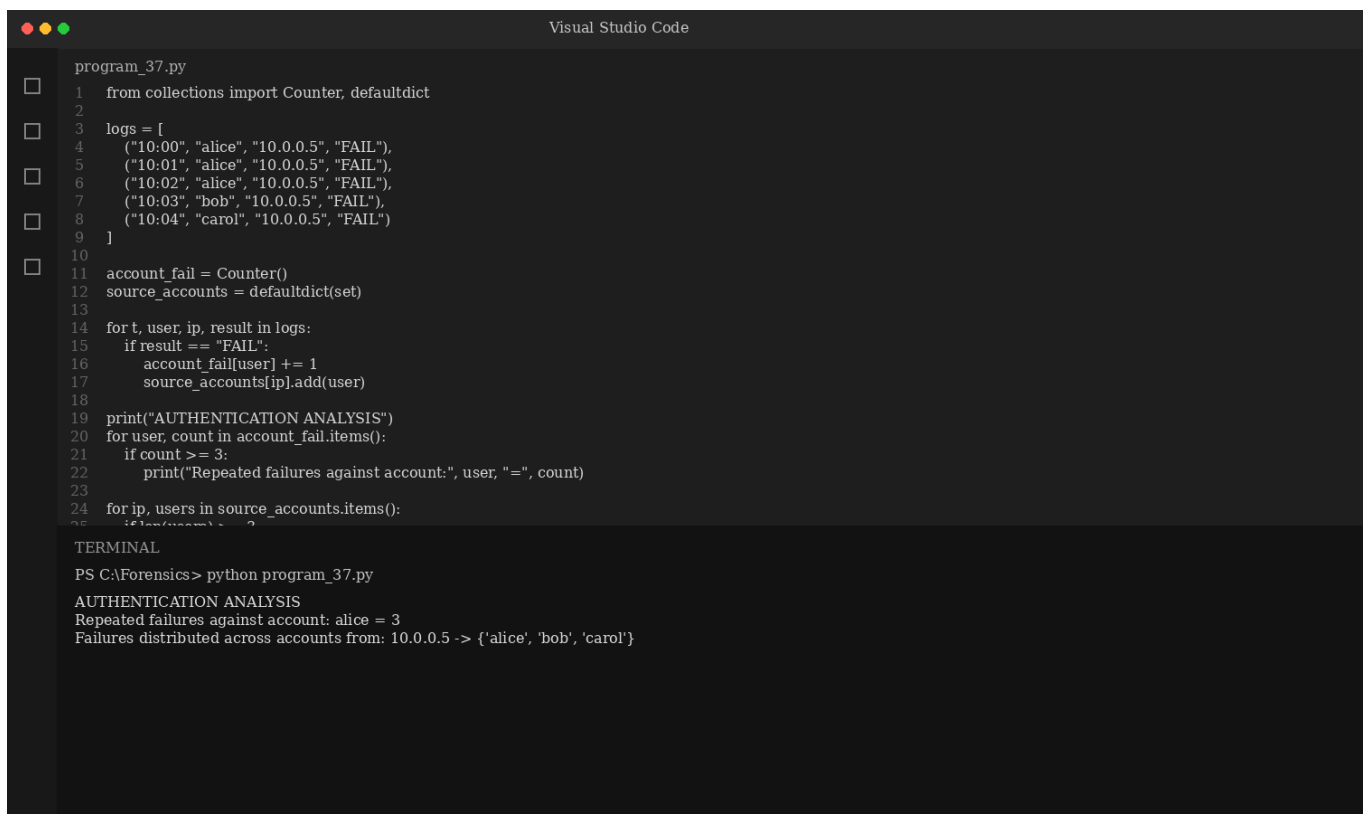
account_fail = Counter()
source_accounts = defaultdict(set)

for t, user, ip, result in logs:
    if result == "FAIL":
        account_fail[user] += 1
        source_accounts[ip].add(user)

print("AUTHENTICATION ANALYSIS")
for user, count in account_fail.items():
    if count >= 3:
        print("Repeated failures against account:", user, "=", count)

for ip, users in source_accounts.items():
    if len(users) >= 3:
        print("Failures distributed across accounts from:", ip, "->", users)
```

OUTPUT



The screenshot shows a Visual Studio Code window with a file named `program_37.py`. The code is a Python script that processes authentication logs. It uses `Counter` from the `collections` module to count failures per user and `defaultdict` to track which users failed from which IP addresses. The logs list five failed attempts: three by 'alice' from '10.0.0.5' and two by 'bob' and 'carol' from the same IP. The script then prints the results of its analysis.

```
program_37.py
1  from collections import Counter, defaultdict
2
3  logs = [
4      ("10:00", "alice", "10.0.0.5", "FAIL"),
5      ("10:01", "alice", "10.0.0.5", "FAIL"),
6      ("10:02", "alice", "10.0.0.5", "FAIL"),
7      ("10:03", "bob", "10.0.0.5", "FAIL"),
8      ("10:04", "carol", "10.0.0.5", "FAIL")
9  ]
10
11 account_fail = Counter()
12 source_accounts = defaultdict(set)
13
14 for t, user, ip, result in logs:
15     if result == "FAIL":
16         account_fail[user] += 1
17         source_accounts[ip].add(user)
18
19 print("AUTHENTICATION ANALYSIS")
20 for user, count in account_fail.items():
21     if count >= 3:
22         print("Repeated failures against account:", user, "=", count)
23
24 for ip, users in source_accounts.items():
25     if len(users) >= 3:
26         print("Failures distributed across accounts from:", ip, "->", users)
```

TERMINAL

```
PS C:\Forensics> python program_37.py
AUTHENTICATION ANALYSIS
Repeated failures against account: alice = 3
Failures distributed across accounts from: 10.0.0.5 -> {'alice', 'bob', 'carol'}
```

RESULT

The program identifies repeated authentication failures and failures distributed across multiple accounts.